

## Exp:1 BFS

### **Program:**

```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])
    visited.add(start)
    print("BFS Traversal:")
    while queue:
        node = queue.popleft()
        print(node, end=" ")
        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['G'],
    'F': [],
    'G': []
}

start = 'A'

bfs(graph, start)
```

## Output:

```
Exp-1program.py - C:\Users\Sakshi\OneDrive\Documents\MLA01 - Lab Exp\Exp - 1(BFS)\Exp-1program.py (3.1...
File Edit Format Run Options Window Help
from collections import deque

# Function to perform Breadth First Search
def bfs(graph, start):
    visited = set()      # Stores visited nodes
    queue = deque([start]) # Queue for BFS

    visited.add(start)

    print("BFS Traversal:")

    while queue:
        node = queue.popleft()
        print(node, end=" ")

        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)

# Graph Representation
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['G'],
    'F': [],
    'G': []
}

# Starting Node
start = 'A'

bfs(graph, start)
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:\Users\Sakshi\OneDrive\Documents\MLA01 - Lab Exp\Exp - 1(BFS)\Exp-1program.
py
BFS Traversal:
A B C D E F G
>>>
```

Ln: 1 Col: 0

Ln: 7 Col: 0